GROUP LAW ON ELLIPTIC CURVES AND ISOGENIES

The divisor-theoretic group law admits a geometric description on a plane cubic. This description has its uses in terms of aiding learning, but it is not typically used beyond that. We will give sketches of all the proofs here, but we will not spend too much time on them.

1. The chord-tangent description

Proposition 1.1. *Let $E \subset \mathbb{P}^2$ be a smooth Weierstrass cubic. Suppose a line L meets E at the points P, Q, R (counted with intersection multiplicity). Then the three points satisfy*

$$P + Q + R = O.$$

in terms of the group law described above Consequently, if R is the third point where the line through P and Q meets E, then

$$P + Q = -R.$$

When $P = Q$, the line through P and Q is interpreted as the tangent line at P.

Proof. Let $\ell(X, Y, Z) = 0$ be an equation for L. Since the line $Z = 0$ meets E at O with multiplicity 3, the rational function ℓ/Z on E has divisor

$$\operatorname{div}\left(\frac{\ell}{Z}\right) = (P) + (Q) + (R) - 3(O) = (P) - (O) + (Q) - (O) + (R) - (O).$$

Hence he preceding corollary gives the result. The tangent case is the same statement with intersection multiplicities taken into account. □

For the general Weierstrass equation, the inverse of an affine point is particularly simple:

$$-(x, y) = (x, -y - a_1x - a_3).$$

Indeed, the vertical line through these two points also passes through O which is the point at infinity.

We record the corresponding algebraic formulas. Let

$$P_1 = (x_1, y_1), \qquad P_2 = (x_2, y_2)$$

be affine points and suppose first that $x_1 \neq x_2$. Put

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}, \qquad \nu = \frac{y_1x_2 - y_2x_1}{x_2 - x_1} = y_1 - \lambda x_1.$$

Then $P_1 + P_2 = (x_3, y_3)$, where

$$\begin{aligned} x_3 &= \lambda^2 + a_1\lambda - a_2 - x_1 - x_2, \\ y_3 &= -(\lambda + a_1)x_3 - \nu - a_3. \end{aligned}$$

For doubling, provided the tangent is not vertical, one uses

$$\lambda = \frac{3x_1^2 + 2a_2x_1 + a_4 - a_1y_1}{2y_1 + a_1x_1 + a_3}, \qquad \nu = y_1 - \lambda x_1,$$

and then the same formulas for x_3 and y_3, with $x_2 = x_1$. If

$$2y_1 + a_1x_1 + a_3 = 0,$$

then the tangent is vertical and $[2]P_1 = O$.

Example 1.2. Suppose $\mathrm{char}(K) \neq 2, 3$ and $E : y^2 = x^3 + Ax + B$. Then

$$-(x, y) = (x, -y).$$

For $P_1 \neq \pm P_2$,

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1},$$

while for doubling,

$$\lambda = \frac{3x_1^2 + A}{2y_1}.$$

In either case,

$$x_3 = \lambda^2 - x_1 - x_2, \qquad y_3 = \lambda(x_1 - x_3) - y_1.$$

These are the familiar addition and doubling formulas used in explicit computations.

1.1. **The group law is algebraic.** So far the construction gives an abelian group law on the points of E. The crucial additional fact is that the group operations themselves are algebraic maps.

Theorem 1.3. *Let E/K be an elliptic curve. The maps*

$$+ : E \times E \longrightarrow E, \qquad (P, Q) \longmapsto P + Q,$$

and

$$- : E \longrightarrow E, \qquad P \longmapsto -P,$$

are morphisms defined over K.

Proof. The negation map is given in Weierstrass coordinates by

$$(x, y) \longmapsto (x, -y - a_1 x - a_3),$$

so it is a rational map $E \dashrightarrow E$. Since E is smooth and the target is projective, every rational map from E to E is regular. Thus negation is a morphism.

For a fixed point $Q \in E$, the formulas above show that translation

$$\tau_Q : E \longrightarrow E, \qquad P \longmapsto P + Q,$$

is a rational map. It is therefore a morphism. Its inverse is τ_{-Q}, so every translation is an automorphism.

The addition formulas give a rational map $E \times E \dashrightarrow E$ which is visibly regular away from the loci where the denominators in the formulas vanish. The source now has dimension 2, so the removable singularity proposition for curves cannot be applied directly. Instead, translate the two variables by auxiliary points Q_1, Q_2 and then translate the answer back:

$$(P_1, P_2) \longmapsto \tau_{-(Q_1+Q_2)}\big(\tau_{Q_1}(P_1) + \tau_{Q_2}(P_2)\big).$$

Wherever the expression is defined, it agrees with $P_1 + P_2$. Given any particular pair (P_1, P_2), the auxiliary points Q_1, Q_2 can be chosen so that none of the exceptional configurations occurs. Thus every point of $E \times E$ has a neighbourhood on which one of these translated formulas is regular. Since the formulas agree on overlaps, they glue to a morphism

$$E \times E \longrightarrow E.$$

All of the formulas and constructions are defined over K. □

Corollary 1.4. *An elliptic curve is a one-dimensional projective abelian algebraic group. In particular:*

(i) every translation τ_Q is an automorphism of E as a curve (but NOT of (E, O) as an elliptic curve, since it does not preserve the identity);

(ii) for every integer n, the multiplication map

$$[n] : E \longrightarrow E, \qquad P \longmapsto [n]P,$$

is a morphism;

(iii) if E is defined over K, then $E(K)$ is an abelian group and the action of G_K on $E(\bar{K})$ is by group automorphisms.

2. Isogenies

Definition 2.1. An *isogeny* $\phi : E_1 \longrightarrow E_2$ is a morphism of curves such that

$$\phi(O_1) = O_2.$$

The constant map with value O_2 is called the *zero isogeny*. An isogeny which is not the zero isogeny is called a *nonzero isogeny*. We say that E_1 and E_2 are *isogenous over* K if there is a nonzero isogeny from E_1 to E_2 defined over K.

Remark 2.2. We follow the convention used in Silverman's book, in which the zero morphism is included among the isogenies. Some authors reserve the word isogeny for a nonconstant morphism preserving the identity. Under that convention, what we call a nonzero isogeny is simply called an isogeny.

A nonzero isogeny is automatically surjective, since a nonconstant morphism between projective curves is surjective.

At first sight, the definition only remembers the identity point. The next theorem shows that this already forces the morphism to preserve addition.

Theorem 2.3. *Let $\phi : E_1 \longrightarrow E_2$ be an isogeny. Then*

$$\phi(P + Q) = \phi(P) + \phi(Q)$$

for all $P, Q \in E_1(\bar{K})$. Thus every isogeny is a homomorphism of abelian groups.

Proof. The assertion is clear for the zero isogeny, so assume that ϕ is nonzero. The maps

$$\kappa_i : E_i \longrightarrow \mathrm{Pic}^0(E_i), \qquad P \longmapsto [(P) - (O_i)]$$

are isomorphisms of abelian groups tautologically.

For every $P \in E_1$, we have

$$\phi_*\big((P) - (O_1)\big) = (\phi(P)) - (O_2).$$

Consequently, the diagram

$$\begin{array}{ccc} E_1 & \xrightarrow{\phi} & E_2 \\ {\scriptstyle \kappa_1}\downarrow & & \downarrow{\scriptstyle \kappa_2} \\ \mathrm{Pic}^0(E_1) & \xrightarrow{\phi_*} & \mathrm{Pic}^0(E_2) \end{array}$$

commutes. Since the vertical maps are group isomorphisms and the bottom map is a group homomorphism, ϕ is a group homomorphism. □

Remark 2.4. There is another proof using a rigidity principle. Consider

$$F : E_1 \times E_1 \longrightarrow E_2, \qquad F(P, Q) = \phi(P + Q) - \phi(P) - \phi(Q).$$

The map F vanishes on both $E_1 \times \{O_1\}$ and $\{O_1\} \times E_1$. A rigidity lemma for morphisms from products of complete varieties then implies that F is identically zero. We will not need this proof here.

2.1. The groups of homomorphisms and rings of endomorphisms.

Definition 2.5. We write

$$\mathrm{Hom}(E_1, E_2)$$

for the set of isogenies $E_1 \to E_2$ defined over $\bar{K}$, including the zero isogeny. We write

$$\mathrm{Hom}_K(E_1, E_2)$$

for the subgroup consisting of the isogenies defined over K.

For $\phi, \psi \in \mathrm{Hom}(E_1, E_2)$, define

$$(\phi + \psi)(P) = \phi(P) + \psi(P).$$

Then $\mathrm{Hom}(E_1, E_2)$ is an abelian group.

The fact that $\phi + \psi$ is again a morphism follows by composing

$$E_1 \xrightarrow{(\phi,\psi)} E_2 \times E_2$$

with the addition morphism $E_2 \times E_2 \to E_2$. It sends O_1 to O_2, and is therefore an isogeny in our convention.

Definition 2.6. The abelian group

$$\mathrm{End}(E) = \mathrm{Hom}(E, E)$$

is called the *endomorphism ring* of E. Addition is defined pointwise and multiplication is composition:

$$\phi\psi = \phi \circ \psi.$$

The identity element for multiplication is $[1]$, and the zero element is the zero isogeny $[0]$.

An invertible element of $\mathrm{End}(E)$ is called an *automorphism of the elliptic curve*. We denote the group of such automorphisms by $\mathrm{Aut}(E)$.